

Week 1 — The piles game (Bulgarian solitaire)

Facilitator guide for the three table adults

The session at a glance

The game. Counters sit in piles. A *move* takes one counter from every pile and puts those counters together as one new pile; piles that reach zero disappear. With 6 counters, $4\ 1\ 1 \rightarrow 3\ 3 \rightarrow 2\ 2\ 2 \rightarrow 3\ 1\ 1\ 1 \rightarrow 4\ 2 \rightarrow 3\ 2\ 1$, and then $3\ 2\ 1 \rightarrow 3\ 2\ 1$ forever: the game *stops*. The theorem behind the session (Brandt, 1982): when the number of counters is a staircase number $1 + 2 + \dots + k$ (3, 6, 10, 15, 21, ...), every start ends at the staircase $k, k - 1, \dots, 1$, which is the only arrangement that a move does not change. For any other number of counters the game never stops; it falls into a loop, and for some numbers (8 and 12 are the smallest) different starts can fall into different loops.

What the children are really investigating, in the order the pages ask it: what always happens (the same move, repeated, must stop or loop); which numbers of counters stop (existence and impossibility of a fixed point); whether every start ends in the same place (completeness: check every arrangement); how many moves it takes (optimality: the slowest start); which arrangements can never be reached (Garden of Eden states). The mathematics section at the end has the proofs; the answer sections have every run written out, so no adult needs to compute anything during the hour.

Shape of the hour: about 4 minutes of free handling of the counters, a 2-minute whole-group launch, about 40 minutes of table work, and 5 minutes of sharing at the end.

Materials and preparation

- Counters, one colour: 15 per child at the K–1 table (45), 15 per child at grades 2–3 (60), 21 per child at grades 4–5 (63), plus 6 large counters for the launch. Also 9 counters of a second colour for the 4–5 table (3 per child), the “extra” counters of Problems 15–16. About 190 in all; more is better, because children like to keep their finished staircases on the table.
- Measure the counters before printing. The K–1 pictures are drawn at a 2 cm pitch (circles 1.8 cm across): counters up to 2 cm across, such as standard 3/4-inch bingo chips, sit one per circle, and “one counter on each circle, none left over” is the whole test in K–1 Problems 2, 4, 6 and 12. One-inch (2.5 cm) two-colour counters overlap on the stairs pictures; with those, have the child build the stairs on the table right next to the picture, tower under tower, and compare, instead of covering the circles.
- Printed packets, one per child: K–1 (7 pages), grades 2–3 (8 pages), grades 4–5 (9 pages). Print one extra of each. Pencils, and ten blank sheets per table: the census problems (2–3 Problem 14, 4–5 Problem 6) are run on blank sheets, and so are extra tries in 4–5 Problem 14.
- This guide for each adult. Read your table’s section before the session; the runs are written out so you can check a child’s record at a glance.
- Playing cards are not needed. If the 4–5 table wants to run a big game fast, a row of cards can stand for the piles (a 5 is a pile of 5, an ace is a pile of 1): a move replaces every card by the next lower card, removes the aces, and adds a card equal to the number of cards there were.
- Preparation: sort the counters into bags of 15 or 21 per child, one bag per child, so that the launch needs no counting out. Put the K–1 children’s chairs on the same side of the table as yours so you can read the page and point.

Adults and tables

A suggestion, not a rule. The parent takes the K–1 table: the job there is reading aloud, pointing, keeping three sets of counters apart, and writing numbers when a child asks; every answer is in the K–1 section below, and the mathematics at that table is carried by the pictures. The second mathematician takes grades 2–3, where the map of all eleven arrangements (Problem 13) is the moment that needs someone who can ask “how do you know there is no other arrow?” The organizer takes grades 4–5, where the arguments (why no arrangement of 8 counters can stand still; why some arrangements can never be reached) are the point. During the share, the two mathematicians should each visit the other tables for a minute.

Free handling (about 4 minutes)

Give every child their bag of counters and nothing else. Let them make piles, towers, rows, and count. Two useful things to say, once, to anyone: “Can you make some piles so that no two piles are the same size?” and “How many counters are in your tallest tower?” Do not explain the move yet.

Whole-group launch (2 minutes)

The organizer stands where everyone can see, with 6 large counters.

1. “Here are 6 counters in three piles: 4, 1 and 1.” Make the piles on the table, visibly.
2. “Here is the only rule of today’s game. A move: I take one counter from every pile,” (do it: three counters in your hand, the two single piles have vanished, the pile of 4 is now 3) “and I put the ones I took together as a new pile.” Put them down: piles of 3 and 3.
3. “Everybody: make piles of 4, 1 and 1 with your own counters. Make one move.” Wait until every table has 3 3. “Make another move.” Everyone should have 2 2 2. Check a few by walking past.
4. “Keep making moves and see what happens. Two questions for the whole hour: does the game ever stop, and does it matter how you start? Your pages tell you what to try.”

Say nothing about staircases, loops or the theorem. The launch is only the move.

K–1 table (three children, one adult)

Read each problem aloud, then read it again in pieces while the child acts. The child places counters on the picture (one counter per circle), makes the move on the table, and draws the result in the next box: one dot per counter and a ring around each pile. If a child does not want to draw, write the pile sizes in the box yourself as the child dictates (“three and one”) and let the child draw the rings around your numbers. Keep each child’s counters on their own page or on a sheet of paper so the sets do not mix.

Order and timing (40 minutes): Problems 1–2 (5 min), 3–5 (11 min), 6–9 (13 min), 10–11 (6 min), 12–13 (10 min; these two, the 10-counter stairs and the 7-counter loop, are the important ones), 14 (for a child who is ahead). That is a little more than 40 minutes of work, on purpose. If the table is behind at 25 minutes, skip Problem 8 (go from 7 to 9) and, if still behind, Problem 5, so that 10–13 are reached.

Two rules for reading the frames, both printed in the problems (stopped in Problem 2, round and round in Problem 3): a game has stopped when a box looks the same as the box before it; a game goes round

and round when the same pictures keep coming back in the same order, that is, when a box looks the same as an earlier box that is not the one just before it.

Answers, K–1

- **Problem 1** (3 counters, piles 1 1 1): $1\ 1\ 1 \rightarrow 3 \rightarrow 2\ 1 \rightarrow 2\ 1$. Boxes 2 and 3 are the same. Say: “The move did not change anything. The game stopped.”
- **Problem 2** (3 counters, one pile): $3 \rightarrow 2\ 1 \rightarrow 2\ 1$. Both starts stopped at the same piles, 2 and 1, and the three counters fit the stairs picture exactly. Ask: “Is that the same place Problem 1 stopped?”
- **Problem 3** (4 counters, one pile): $4 \rightarrow 3\ 1 \rightarrow 2\ 2 \rightarrow 2\ 1\ 1 \rightarrow 3\ 1 \rightarrow 2\ 2$. Box 4 is the same as box 1, box 5 the same as box 2. It never stops: the pictures 3 1, 2 2, 2 1 1 keep coming back in that order. The child should make a few more moves and say “it goes round and round.”
- **Problem 4** (4 counters, piles 2 2): $2\ 2 \rightarrow 2\ 1\ 1 \rightarrow 3\ 1 \rightarrow 2\ 2$. Box 3 is the same as the start. Four counters do not fit either stairs picture exactly: one counter is left over on the 2–1 stairs, and two circles stay empty on the 3–2–1 stairs. This is the first impossibility of the day: there is no stairs with 4 counters.
- **Problem 5** (5 counters, one pile): $5 \rightarrow 4\ 1 \rightarrow 3\ 2 \rightarrow 2\ 2\ 1 \rightarrow 3\ 1\ 1 \rightarrow 3\ 2$. Boxes 2 and 5 are the same; the game goes round on the three pictures 3 2, 2 2 1, 3 1 1 and never stops.
- **Problem 6** (6 counters, one pile): $6 \rightarrow 5\ 1 \rightarrow 4\ 2 \rightarrow 3\ 2\ 1 \rightarrow 3\ 2\ 1$. It stops at 3 2 1 after 3 moves (3 goes in the circle: the number of the first box that shows the stairs), and the counters fit the 3–2–1 stairs exactly.
- **Problem 7** (6 counters, six piles of 1): $1\ 1\ 1\ 1\ 1\ 1 \rightarrow 6 \rightarrow 5\ 1 \rightarrow 4\ 2 \rightarrow 3\ 2\ 1 \rightarrow 3\ 2\ 1$. Four moves to reach the stairs. Notice with the child that after one move this is Problem 6 again.
- **Problem 8** (6 counters, piles 3 3): $3\ 3 \rightarrow 2\ 2\ 2 \rightarrow 3\ 1\ 1\ 1 \rightarrow 4\ 2 \rightarrow 3\ 2\ 1 \rightarrow 3\ 2\ 1$. Four moves.
- **Problem 9** (6 counters, piles 2 2 1 1): $2\ 2\ 1\ 1 \rightarrow 4\ 1\ 1 \rightarrow 3\ 3 \rightarrow 2\ 2\ 2 \rightarrow 3\ 1\ 1\ 1 \rightarrow 4\ 2 \rightarrow 3\ 2\ 1 \rightarrow 3\ 2\ 1$. Six moves, the most of the four games (3, 4, 4, 6). All four games with 6 counters stopped at the same stairs. Ask: “Is there any way to start 6 counters so that it does not end at the stairs?” Let them try one or two of their own; every start with 6 counters ends at 3 2 1 (the full list is in the tables at the end of this guide).
- **Problem 10** (table): 3 stopped; 4 round and round; 5 round and round; 6 stopped; later, 7 round and round and 10 stopped.
- **Problem 11** (build the stairs): the stairs with 4 steps use 10 counters; with 5 steps, 15. Each child has 15 counters, which is why the problem says to take each stairs apart before building the next (the four together would need 34). A child who counts 3, 6, 10, 15 aloud has found the staircase numbers. Ask: “How many more counters did the bigger stairs need than the smaller one?” (3, then 4, then 5.) For the share, build the 3-, 6- and 10-stairs side by side from the table’s pooled counters (19 of the 45).
- **Problem 12** (10 counters, one pile): $10 \rightarrow 9\ 1 \rightarrow 8\ 2 \rightarrow 7\ 2\ 1 \rightarrow 6\ 3\ 1 \rightarrow 5\ 3\ 2 \rightarrow 4\ 3\ 2\ 1 \rightarrow 4\ 3\ 2\ 1$. Six moves; it stops at the 4–3–2–1 stairs and the counters fit the big picture exactly.
- **Problem 13** (7 counters, one pile): $7 \rightarrow 6\ 1 \rightarrow 5\ 2 \rightarrow 4\ 2\ 1 \rightarrow 3\ 3\ 1 \rightarrow 3\ 2\ 2 \rightarrow 3\ 2\ 1\ 1 \rightarrow 4\ 2\ 1$. Box 7 is the same as box 3; the game goes round on four pictures and never stops.
- **Problem 14** (8 counters, piles 2 2 2 2): $2\ 2\ 2\ 2 \rightarrow 4\ 1\ 1\ 1\ 1 \rightarrow 5\ 3 \rightarrow 4\ 2\ 2 \rightarrow 3\ 3\ 1\ 1 \rightarrow 4\ 2\ 2$. Boxes 3 and 5 are the same; this game goes round on only two pictures, 4 2 2 and 3 3 1 1, so 2 goes in the circle (the game with 7 counters went round on four).

Hints, K–1, in the order to give them

1. A child takes counters from only some piles: “One from every pile. Point to each pile and take one.” Have the child put the taken counters in their other hand before making the new pile.
2. A child forgets that the counters taken form one pile: “All the ones in your hand go down together.”
3. A child cannot see whether two boxes are the same: count the piles in each box, then the counters in the biggest pile. Say the sizes aloud, “three and one”, for both boxes.
4. Round and round is hard to see: put a finger on box 1 and ask “where is this picture again?”
5. For the stairs test (Problems 2, 4, 6, 12): put the piles down as towers, biggest on the left, next to the picture, before placing counters on the circles. In Problem 4 the point is that 4 counters fit neither stairs picture: say “one left over” and “two circles empty” with the child.

If the K–1 table stalls or finishes early

Stalls: skip to Problem 6 (the 6-counter game from one pile is short and satisfying), then come back. If drawing is slowing everything, you draw, the child moves. If a child only wants to build, let them build the stairs of Problem 11 and count. If the table is behind at 25 minutes, skip Problem 8 and then Problem 5, as in the timing above.

Finishes early, in this order: (1) 8 counters in one pile: $8 \rightarrow 7\ 1 \rightarrow 6\ 2 \rightarrow 5\ 2\ 1 \rightarrow 4\ 3\ 1 \rightarrow 3\ 3\ 2 \rightarrow 3\ 2\ 2\ 1 \rightarrow 4\ 2\ 1\ 1 \rightarrow 4\ 3\ 1$; it goes round on four pictures, so with 8 counters the two starts of Problems 14 and this one go round on different pictures. (2) “Which numbers stopped?” 3, 6, 10: build all three stairs in a row and ask what the next one needs (15). (3) 15 counters as five piles of 3: $3\ 3\ 3\ 3\ 3 \rightarrow 5\ 2\ 2\ 2\ 2\ 2 \rightarrow 6\ 4\ 1\ 1\ 1\ 1\ 1 \rightarrow 7\ 5\ 3 \rightarrow 6\ 4\ 3\ 2 \rightarrow 5\ 4\ 3\ 2\ 1$, which stops after 5 moves at the 5-step stairs. Any of these can be done on a blank sheet with boxes you draw.

Grades 2–3 table (four children, one adult)

Each child keeps their own counters and their own packet, but let pairs check each other’s rows: a wrong row is usually a pile that was not reduced. In the tables, the children write pile sizes biggest first, separated by spaces, so that equal arrangements look equal. The children made the first two moves of Problem 1 during the launch; Problem 1 is where they write them down. Have them rebuild 4 1 1 and record from the start.

Order and timing (40 minutes): Problems 1–4 (12 min), 5–8 (13 min), 9–10 (6 min), 11 (3 min), 12–13 (10 min). That is about 44 minutes of work. Problems 14 and 15 are the finishing-early work: offer 15 (3 min) first, then 14 shared out across the table (10 min). If the table is behind at 25 minutes, skip Problem 11; Problems 12 and 13 (the two loops for 8 counters, and the map) must be reached. If the table is slow from the start, Problem 3 can be split: two children run 3 3 and two run 2 2 1 1, and they pool the results in Problem 4.

Answers, grades 2–3

- **Problem 1** (4 1 1): $4\ 1\ 1 \rightarrow 3\ 3 \rightarrow 2\ 2\ 2 \rightarrow 3\ 1\ 1\ 1 \rightarrow 4\ 2 \rightarrow 3\ 2\ 1 \rightarrow 3\ 2\ 1$. The row for move 6 repeats move 5, so the game stopped after 5 moves at 3 2 1.
- **Problem 2**: $6 \rightarrow 5\ 1 \rightarrow 4\ 2 \rightarrow 3\ 2\ 1 \rightarrow 3\ 2\ 1$ (3 moves). $1\ 1\ 1\ 1\ 1\ 1 \rightarrow 6 \rightarrow 5\ 1 \rightarrow 4\ 2 \rightarrow 3\ 2\ 1 \rightarrow 3\ 2\ 1$ (4 moves).

- **Problem 3:** $3\ 3 \rightarrow 2\ 2\ 2 \rightarrow 3\ 1\ 1\ 1 \rightarrow 4\ 2 \rightarrow 3\ 2\ 1 \rightarrow 3\ 2\ 1$ (4 moves). $2\ 2\ 1\ 1 \rightarrow 4\ 1\ 1 \rightarrow 3\ 3 \rightarrow 2\ 2\ 2 \rightarrow 3\ 1\ 1\ 1 \rightarrow 4\ 2 \rightarrow 3\ 2\ 1 \rightarrow 3\ 2\ 1$ (6 moves).
- **Problem 4:** moves until it stopped: $4\ 1\ 1$ takes 5; 6 takes 3; $1\ 1\ 1\ 1\ 1\ 1$ takes 4; $3\ 3$ takes 4; $2\ 2\ 1\ 1$ takes 6, the longest. All five end at $3\ 2\ 1$, which drawn as towers is a staircase. Ask “did any start end somewhere else?” and then “could one?”; hold the second question until Problem 13.
- **Problem 5 (4):** $4 \rightarrow 3\ 1 \rightarrow 2\ 2 \rightarrow 2\ 1\ 1 \rightarrow 3\ 1 \rightarrow 2\ 2 \rightarrow 2\ 1\ 1$. Rows 4 (and 1) are $3\ 1$; the game never stops. It cycles through $3\ 1, 2\ 2, 2\ 1\ 1$.
- **Problem 6** (all five arrangements of 4): $A\ (4) \rightarrow D$; $B\ (2\ 2) \rightarrow E$; $C\ (1\ 1\ 1\ 1) \rightarrow A$; $D\ (3\ 1) \rightarrow B$; $E\ (2\ 1\ 1) \rightarrow D$. The game goes round on D, B, E . The arrangement never reached by a move is C , four piles of 1: after any move there is a pile of size equal to the number of piles before the move, which is never 1 unless there was only one pile, and one pile of 4 becomes $3\ 1$, not $1\ 1\ 1\ 1$.
- **Problem 7** (5 and $1\ 1\ 1\ 1\ 1$): $5 \rightarrow 4\ 1 \rightarrow 3\ 2 \rightarrow 2\ 2\ 1 \rightarrow 3\ 1\ 1 \rightarrow 3\ 2 \rightarrow 2\ 2\ 1$; $1\ 1\ 1\ 1\ 1 \rightarrow 5 \rightarrow 4\ 1 \rightarrow 3\ 2 \rightarrow 2\ 2\ 1 \rightarrow 3\ 1\ 1 \rightarrow 3\ 2$. Yes, both go round on the same three arrangements $3\ 2, 2\ 2\ 1, 3\ 1\ 1$; the second game is the first one delayed by one move.
- **Problem 8** (7 and $2\ 2\ 2\ 1$): $7 \rightarrow 6\ 1 \rightarrow 5\ 2 \rightarrow 4\ 2\ 1 \rightarrow 3\ 3\ 1 \rightarrow 3\ 2\ 2 \rightarrow 3\ 2\ 1\ 1 \rightarrow 4\ 2\ 1 \rightarrow 3\ 3\ 1$. Rows 7 and 8 repeat rows 3 and 4. $2\ 2\ 2\ 1 \rightarrow 4\ 1\ 1\ 1 \rightarrow 4\ 3 \rightarrow 3\ 2\ 2 \rightarrow 3\ 2\ 1\ 1 \rightarrow 4\ 2\ 1 \rightarrow 3\ 3\ 1 \rightarrow 3\ 2\ 2$; row 7 repeats row 3. Yes, both games go round on the same four arrangements, $4\ 2\ 1, 3\ 3\ 1, 3\ 2\ 2, 3\ 2\ 1\ 1$, entered at different places. So far, every number of counters has had one place where every start ends; Problem 12 breaks that.
- **Problem 9** (run and table): $1\ 1\ 1 \rightarrow 3 \rightarrow 2\ 1 \rightarrow 2\ 1$, so 3 stops after 2 moves, at $2\ 1$. Table: 3 stops; 4 goes round, 3 arrangements; 5 goes round, 3 arrangements; 6 stops; 7 goes round, 4 arrangements.
- **Problem 10:** the stairs $3\ 2\ 1$ with a tower of 4 added is $4\ 3\ 2\ 1$, 10 counters (each child has 15, enough). Prediction: most children say “stops” because 3 and 6 stopped; accept either prediction and test. $10 \rightarrow 9\ 1 \rightarrow 8\ 2 \rightarrow 7\ 2\ 1 \rightarrow 6\ 3\ 1 \rightarrow 5\ 3\ 2 \rightarrow 4\ 3\ 2\ 1 \rightarrow 4\ 3\ 2\ 1$: stops after 6 moves at the 4–3–2–1 stairs.
- **Problem 11:** $2\ 2\ 2\ 2\ 2 \rightarrow 5\ 1\ 1\ 1\ 1\ 1 \rightarrow 6\ 4 \rightarrow 5\ 3\ 2 \rightarrow 4\ 3\ 2\ 1 \rightarrow 4\ 3\ 2\ 1$ (4 moves). $5\ 5 \rightarrow 4\ 4\ 2 \rightarrow 3\ 3\ 3\ 1 \rightarrow 4\ 2\ 2\ 2 \rightarrow 4\ 3\ 1\ 1\ 1 \rightarrow 5\ 3\ 2 \rightarrow 4\ 3\ 2\ 1 \rightarrow 4\ 3\ 2\ 1$ (6 moves). Both end at the stairs.
- **Problem 12** (8 counters): $8 \rightarrow 7\ 1 \rightarrow 6\ 2 \rightarrow 5\ 2\ 1 \rightarrow 4\ 3\ 1 \rightarrow 3\ 3\ 2 \rightarrow 3\ 2\ 2\ 1 \rightarrow 4\ 2\ 1\ 1 \rightarrow 4\ 3\ 1$, a loop of four ($4\ 3\ 1, 3\ 3\ 2, 3\ 2\ 2\ 1, 4\ 2\ 1\ 1$). $2\ 2\ 2\ 2 \rightarrow 4\ 1\ 1\ 1\ 1 \rightarrow 5\ 3 \rightarrow 4\ 2\ 2 \rightarrow 3\ 3\ 1\ 1 \rightarrow 4\ 2\ 2$, a loop of two ($4\ 2\ 2, 3\ 3\ 1\ 1$). No: the two games go round on different arrangements. This is the first time in the session that the start matters for where you end up.
- **Problem 13** (the map of 6 counters). Arrows: $A\ (2\ 2\ 1\ 1) \rightarrow B\ (4\ 1\ 1) \rightarrow C\ (3\ 3) \rightarrow D\ (2\ 2\ 2) \rightarrow I\ (3\ 1\ 1\ 1) \rightarrow H\ (4\ 2) \rightarrow K\ (3\ 2\ 1)$; $E\ (1\ 1\ 1\ 1\ 1\ 1) \rightarrow F\ (6) \rightarrow G\ (5\ 1) \rightarrow H$; $J\ (2\ 1\ 1\ 1\ 1) \rightarrow G$; $K \rightarrow K$ (the small loop arrow the problem asks for). The cells are laid out so that every arrow goes to a touching or diagonal neighbour and no two arrows cross: the top row $A\ B\ C\ D$ reads left to right, D drops to I , I goes down-left to H , H goes right to K ; the middle row $E\ F\ G$ reads left to right and G drops to H ; J goes up-right to G . No arrow points at A, E or J . Every path ends at K . From A it is 6 arrows to K . Two arrows arrive at G (from F and J) and two at H (from G and I). Each picture has exactly one arrow leaving it, because the move leaves no choice; that is why the drawing is a tree of paths flowing into K , not a tangle. Since these are all the arrangements of 6 counters, the map is a complete proof that every start with 6 counters ends at the stairs.
- **Problem 14** (all 22 arrangements of 8): the seven that go round on two are $6\ 1\ 1, 5\ 3, 4\ 2\ 2, 4\ 1\ 1\ 1, 3\ 3\ 1\ 1, 2\ 2\ 2\ 2, 2\ 2\ 1\ 1\ 1\ 1$. The other fifteen go round on four. The stopping rule printed in the

problem works because every start reaches one of the six loop arrangements within 5 moves (the slowest is 1 1 1 1 1 1 1: 8 after one move, then 7 1, 6 2, 5 2 1, 4 3 1); a child who has made 6 moves without reaching one has a wrong row, usually a forgotten pile. Every run is written out in the table for 8 counters at the end of this guide.

- **Problem 15:** the 5-step stairs has 15 counters. 3 3 3 3 3 → 5 2 2 2 2 2 → 6 4 1 1 1 1 1 → 7 5 3 → 6 4 3 2 → 5 4 3 2 1 → 5 4 3 2 1: 5 moves.

Hints, grades 2–3, in the order to give them

1. Wrong row: “Count your piles before the move. That number is the size of the new pile.” Then “every old pile lost one.”
2. A child does not notice a repeat: “Read row 1 aloud. Now find it lower down.”
3. Problem 6: “Make picture A with counters, make the move, then look for the picture you have.”
4. Problem 10: “Line the stairs up: 2 1, then 3 2 1, then 4 3 2 1. How many counters does each one add?” (3, then 4; the next stairs adds 5, which is Problem 15.)
5. Problem 13: “Do the picture with the most piles first (E). Where does its arrow go?” Then “which pictures have you not drawn an arrow from yet?” Finally “are there pictures with no arrow coming in?”
6. After Problem 13, for the question “could a start with 6 counters end somewhere else?”: “Every arrangement of 6 counters is on this map. Where does each arrow chain end?”

If the grades 2–3 table stalls or finishes early

Stalls: go back to counters and one move at a time, one child moving while the others predict the next row aloud; then continue the table. If Problem 13 is too much, do it as a group with one map: each child takes three pictures. If the table is far behind at 25 minutes, skip Problems 11 and 12 and go to the map.

Finishes early, in this order: (1) “Find a start with 6 counters that takes more than 6 moves” (there is none; the map shows 6 is the most, from A). (2) “Predict the loop for 11 counters”: the loop for 7 counters is the 3–2–1 stairs with one extra counter placed on a step or as a new pile of 1 (4 2 1, 3 3 1, 3 2 2, 3 2 1 1), so for 11 counters guess the 4–3–2–1 stairs plus one extra: 5 3 2 1, 4 4 2 1, 4 3 3 1, 4 3 2 2, 4 3 2 1 1; running 5 3 2 1 → 4 4 2 1 → 4 3 3 1 → 4 3 2 2 → 4 3 2 1 1 → 5 3 2 1 confirms it. (3) 15 counters from one pile: 15 → 14 1 → 13 2 → 12 2 1 → 11 3 1 → 10 3 2 → 9 3 2 1 → 8 4 2 1 → 7 4 3 1 → 6 4 3 2 → 5 4 3 2 1, 10 moves.

Grades 4–5 table (three children, one adult)

This table runs the same experiments and then argues. Let the children run games in parallel and pool results; insist that rows are written biggest first, and that “the loop” is named by listing its arrangements.

Order and timing (40 minutes): Problems 1–3 (11 min), 4–6 (13 min), 7–9 (14 min), 10–13 (19 min, the arguments), 14–17 (whatever is left, in any order the children like; 14 and 15 are the best). That is more than 40 minutes on purpose, and the arguments must not be lost at the end: if the table is behind at 25 minutes, cut Problem 6 to the seven S starts (give them the list and ask them to check it) and Problems 8 and 9 to one run each (keep 10 from one pile and 5 4, and fill the table from the runs they have), so that Problems 10, 12 and 13 are reached. Problems 14–17 are for a table that finishes.

Answers, grades 4–5

- **Problem 1:** $2\ 2\ 1\ 1 \rightarrow 4\ 1\ 1 \rightarrow 3\ 3 \rightarrow 2\ 2\ 2 \rightarrow 3\ 1\ 1\ 1 \rightarrow 4\ 2 \rightarrow 3\ 2\ 1 \rightarrow 3\ 2\ 1$; 6 moves.
- **Problem 2:** $3\ 3 \rightarrow 2\ 2\ 2 \rightarrow 3\ 1\ 1\ 1 \rightarrow 4\ 2 \rightarrow 3\ 2\ 1$ (4 moves); $6 \rightarrow 5\ 1 \rightarrow 4\ 2 \rightarrow 3\ 2\ 1$ (3); $1\ 1\ 1\ 1\ 1\ 1 \rightarrow 6 \rightarrow \dots$ (4). All end at $3\ 2\ 1$.
- **Problem 3** (moves until it stops, for all 11 arrangements of 6, the number of moves in brackets): 6 (3), $5\ 1$ (2), $4\ 2$ (1), $4\ 1\ 1$ (5), $3\ 3$ (4), $3\ 2\ 1$ (0), $3\ 1\ 1\ 1$ (2), $2\ 2\ 2$ (3), $2\ 2\ 1\ 1$ (6), $2\ 1\ 1\ 1\ 1$ (3), $1\ 1\ 1\ 1\ 1\ 1$ (4). Since these are all the arrangements, this list proves that every start with 6 counters ends at the stairs, and that 6 moves is the most it can take. Say this explicitly: a complete list is a proof.
- **Problem 4:** $8 \rightarrow 7\ 1 \rightarrow 6\ 2 \rightarrow 5\ 2\ 1 \rightarrow 4\ 3\ 1 \rightarrow 3\ 3\ 2 \rightarrow 3\ 2\ 2\ 1 \rightarrow 4\ 2\ 1\ 1 \rightarrow 4\ 3\ 1$. Row 8 repeats row 4; the loop is $4\ 3\ 1, 3\ 3\ 2, 3\ 2\ 2\ 1, 4\ 2\ 1\ 1$: four arrangements.
- **Problem 5:** $2\ 2\ 2\ 2 \rightarrow 4\ 1\ 1\ 1\ 1 \rightarrow 5\ 3 \rightarrow 4\ 2\ 2 \rightarrow 3\ 3\ 1\ 1 \rightarrow 4\ 2\ 2$: a loop of two, $4\ 2\ 2$ and $3\ 3\ 1\ 1$. $5\ 1\ 1\ 1 \rightarrow 4\ 4 \rightarrow 3\ 3\ 2 \rightarrow 3\ 2\ 2\ 1 \rightarrow 4\ 2\ 1\ 1 \rightarrow 4\ 3\ 1 \rightarrow 3\ 3\ 2$: the same loop of four as the game from 8. So the games from 8 and from $5\ 1\ 1\ 1$ share a loop; the game from $2\ 2\ 2\ 2$ does not.
- **Problem 6:** S (the short loop): $6\ 1\ 1, 5\ 3, 4\ 2\ 2, 4\ 1\ 1\ 1\ 1, 3\ 3\ 1\ 1, 2\ 2\ 2\ 2, 2\ 2\ 1\ 1\ 1\ 1$, seven arrangements. L (the long loop): the other fifteen. No N: no start ends in a new loop, so for 8 counters there are exactly two loops, and this census is the proof. Every start reaches a loop arrangement within 5 moves, so the “stop when you reach an arrangement already seen” rule ends every run quickly; a run that is still going after 6 moves has a wrong row. Each run is in the 8-counter table at the end.
- **Problem 7:** $4 \rightarrow 3\ 1 \rightarrow 2\ 2 \rightarrow 2\ 1\ 1 \rightarrow 3\ 1$ (row 4 repeats row 1). $5 \rightarrow 4\ 1 \rightarrow 3\ 2 \rightarrow 2\ 2\ 1 \rightarrow 3\ 1\ 1 \rightarrow 3\ 2$ (row 5 repeats row 2). The seven arrangements of 5: $5; 4\ 1; 3\ 2; 3\ 1\ 1; 2\ 2\ 1; 2\ 1\ 1\ 1; 1\ 1\ 1\ 1\ 1$. The argument: the rows Start, 1, 2, $\dots$, 7 are eight arrangements, but only seven exist, so two rows are equal, and the later of the two is a repeat, at move 7 at the latest. Then push: once a row repeats an earlier row, everything after it repeats too, because each row is determined by the row above it. That is why every game, with any number of counters, either stops or goes round.
- **Problem 8:** $10 \rightarrow 9\ 1 \rightarrow 8\ 2 \rightarrow 7\ 2\ 1 \rightarrow 6\ 3\ 1 \rightarrow 5\ 3\ 2 \rightarrow 4\ 3\ 2\ 1$ (6 moves); $5\ 5 \rightarrow 4\ 4\ 2 \rightarrow 3\ 3\ 3\ 1 \rightarrow 4\ 2\ 2\ 2 \rightarrow 4\ 3\ 1\ 1\ 1 \rightarrow 5\ 3\ 2 \rightarrow 4\ 3\ 2\ 1$ (6); $2\ 2\ 2\ 2\ 2 \rightarrow 5\ 1\ 1\ 1\ 1\ 1 \rightarrow 6\ 4 \rightarrow 5\ 3\ 2 \rightarrow 4\ 3\ 2\ 1$ (4).
- **Problem 9:** $1\ 1\ 1 \rightarrow 3 \rightarrow 2\ 1 \rightarrow 2\ 1$. $7 \rightarrow 6\ 1 \rightarrow 5\ 2 \rightarrow 4\ 2\ 1 \rightarrow 3\ 3\ 1 \rightarrow 3\ 2\ 2 \rightarrow 3\ 2\ 1\ 1 \rightarrow 4\ 2\ 1$ (row 7 repeats row 3; loop of four). $5\ 4 \rightarrow 4\ 3\ 2 \rightarrow 3\ 3\ 2\ 1 \rightarrow 4\ 2\ 2\ 1 \rightarrow 4\ 3\ 1\ 1 \rightarrow 4\ 3\ 2$ (row 5 repeats row 1; loop of four: $4\ 3\ 2, 3\ 3\ 2\ 1, 4\ 2\ 2\ 1, 4\ 3\ 1\ 1$). Table: 3 stops; 4 goes round, one loop of 3; 5 goes round, one loop of 3; 6 stops; 7 goes round, a loop of 4; 8 goes round, loops of 4 and of 2; 9 goes round, a loop of 4; 10 stops. For 7 and 9 the children have seen one loop each; in fact it is the only one (all 15 arrangements of 7 and all 30 of 9 reach it; tables at the end). The stopping numbers are 3, 6, 10.
- **Problem 10:** $4\ 3\ 2\ 1$ does not change, and it is the only arrangement of 10 counters that does not. There is no such arrangement of 8 counters. The argument, in steps the children can follow: (a) If the move changes nothing, the biggest pile after the move is the same size as the biggest pile before. Every old pile lost a counter, so none of them can be that big any more; only the new pile can. So the new pile is the biggest pile, and its size is the number of piles. (b) Apart from the new pile, what you see after the move is every old pile with one counter taken away, including the old biggest pile. That must be the same collection as the old piles without the biggest one. Match them up from the top: the second biggest pile is one less than the biggest, the third is one less than the second, and so on, down to a pile of 1 that vanished. So the piles must be $k, k-1, \dots, 1$, a staircase. (c) A staircase has

$1 + 2 + \cdots + k$ counters, and 8 is not such a number (6 is too small, 10 too big). So no arrangement of 8 counters can stand still.

- **Problem 11:** 15, 21, 28; stairs 5–4–3–2–1, 6–5–4–3–2–1, 7–6–5–4–3–2–1. $3\ 3\ 3\ 3\ 3 \rightarrow 5\ 2\ 2\ 2\ 2\ 2 \rightarrow 6\ 4\ 1\ 1\ 1\ 1\ 1 \rightarrow 7\ 5\ 3 \rightarrow 6\ 4\ 3\ 2 \rightarrow 5\ 4\ 3\ 2\ 1 \rightarrow 5\ 4\ 3\ 2\ 1$: stops after 5 moves.
- **Problem 12** (undoing a move). The method: the new pile after a move equals the number of piles before it; so choose which pile is “new”, give every other pile one counter back, and add piles of 1 until the number of piles equals the new pile. Into 3 2 1: from 4 2 (new pile 2: the other piles 3 1 came from 4 2, and 2 piles is right) and from 3 2 1 itself (new pile 3: others 2 1 came from 3 2, plus one pile of 1). Into 5 3 2: from 6 4 (new pile 2), from 6 3 1 (new pile 3: others 5 2 came from 6 3, plus one pile of 1), from 4 3 1 1 1 (new pile 5: others 3 2 came from 4 3, plus three piles of 1). Into 2 2 2 2: nothing, and the “why not” blank wants this reason: the new pile would be a 2, so there were only 2 piles before the move, but three other piles are still there afterwards and each of them was there before; impossible. (Problem 13 turns this into a general test.)
- **Problem 13:** 4 4 is reached from 5 1 1 1; 3 3 1 1 from 4 2 2; 5 1 1 1 from 2 2 2 1 1; 4 2 1 1 from 3 2 2 1. Unreachable: 2 2 2 2, 2 2 2 1 1, 3 2 1 1 1, 3 1 1 1 1 1. What they have in common: the biggest pile is smaller than the number of piles minus one. The reason: after a move, the new pile is as big as the number of piles before the move, and that number is at least the number of piles after the move minus one (all the piles you see afterwards, except the new one, were there before). So after any move, the biggest pile is at least the number of piles minus one. Test: 6 1 1 has biggest pile 6 and 3 piles, so it can be reached (from 2 2 1 1 1 1, new pile 6); 2 2 2 1 1 1 1 has biggest pile 2 and 7 piles, so it cannot. Arrangements that cannot be reached are called Garden of Eden states; they are marked in the tables at the end.
- **Problem 14:** the slowest starts with 10 counters take 12 moves: 3 3 2 1 1, 3 2 2 2 1, 3 2 2 1 1 1. For example $3\ 3\ 2\ 1\ 1 \rightarrow 5\ 2\ 2\ 1 \rightarrow 4\ 4\ 1\ 1 \rightarrow 4\ 3\ 3 \rightarrow 3\ 3\ 2\ 2 \rightarrow 4\ 2\ 2\ 1\ 1 \rightarrow 5\ 3\ 1\ 1 \rightarrow 4\ 4\ 2 \rightarrow 3\ 3\ 3\ 1 \rightarrow 4\ 2\ 2\ 2 \rightarrow 4\ 3\ 1\ 1\ 1 \rightarrow 5\ 3\ 2 \rightarrow 4\ 3\ 2\ 1$. Eleven moves: 6 2 1 1, 5 2 2 1, 5 2 1 1 1. The number of moves for all 42 arrangements is in the table at the end. A theorem (Igusa 1985, Etienne 1991) says that with $1 + 2 + \cdots + k$ counters no start needs more than $k^2 - k$ moves: 12 for 10 counters, 20 for 15, 30 for 21.
- **Problem 15:** with the second-colour counter on the stairs 3 2 1: 4 2 1 is the extra on the tower of 3; 3 3 1 on the tower of 2; 3 2 2 on the tower of 1; 3 2 1 1 is the extra as a new pile of 1. Following the loop in order, the extra counter steps down one tower per move (3, 2, 1, new pile) and then reappears on top; let the children run the loop with the coloured counter on the table and watch it. The five arrangements are 5 3 2 1, 4 4 2 1, 4 3 3 1, 4 3 2 2, 4 3 2 1 1, and $5\ 3\ 2\ 1 \rightarrow 4\ 4\ 2\ 1 \rightarrow 4\ 3\ 3\ 1 \rightarrow 4\ 3\ 2\ 2 \rightarrow 4\ 3\ 2\ 1\ 1 \rightarrow 5\ 3\ 2\ 1$ stays inside the list: the extra counter walks down the staircase one step per move and then reappears on top. For 8 counters, the six arrangements with one extra counter on each of two different steps of 3–2–1 are 4 3 1, 4 2 2, 4 2 1 1, 3 3 2, 3 3 1 1, 3 2 2 1, which are exactly the four arrangements of the long loop together with the two of the short loop. In the long loop the two extras sit on neighbouring steps and walk down together; in the short loop they sit two steps apart.
- **Problem 16:** the ten arrangements are 5 4 2 1, 5 3 3 1, 5 3 2 2, 5 3 2 1 1, 4 4 3 1, 4 4 2 2, 4 4 2 1 1, 4 3 3 2, 4 3 3 1 1, 4 3 2 2 1. The prediction to hope for, by analogy with 8 counters: two loops, one for “extras on neighbouring steps” (5 4 2 1, 4 4 3 1, 4 3 3 2, 4 3 2 2 1, 5 3 2 1 1) and one for “extras two steps apart” (5 3 3 1, 4 4 2 2, 4 3 3 1 1, 5 3 2 2, 4 4 2 1 1); accept any prediction with a reason and test it. Two loops of five: $5\ 4\ 2\ 1 \rightarrow 4\ 4\ 3\ 1 \rightarrow 4\ 3\ 3\ 2 \rightarrow 4\ 3\ 2\ 2\ 1 \rightarrow 5\ 3\ 2\ 1\ 1 \rightarrow 5\ 4\ 2\ 1$ (extras on neighbouring steps) and $5\ 3\ 3\ 1 \rightarrow 4\ 4\ 2\ 2 \rightarrow 4\ 3\ 3\ 1\ 1 \rightarrow 5\ 3\ 2\ 2 \rightarrow 4\ 4\ 2\ 1\ 1 \rightarrow 5\ 3\ 3\ 1$ (extras two steps apart). Not the same loop. Starts outside the list that end in the second loop: 6 4 2 (one move), 7 5 (two), 3 3 3

3 (four), 2 2 2 2 2 2 (four), 4 4 4 (five: $4\ 4\ 4 \rightarrow 3\ 3\ 3\ 3 \rightarrow 4\ 2\ 2\ 2\ 2 \rightarrow 5\ 3\ 1\ 1\ 1\ 1 \rightarrow 6\ 4\ 2 \rightarrow 5\ 3\ 3\ 1$). Of the 77 arrangements of 12 counters, 45 end in the first loop and 32 in the second.

- **Problem 17:** from 5 5 5 it is 17 moves: $5\ 5\ 5 \rightarrow 4\ 4\ 4\ 3 \rightarrow 4\ 3\ 3\ 3\ 2 \rightarrow 5\ 3\ 2\ 2\ 2\ 1 \rightarrow 6\ 4\ 2\ 1\ 1\ 1 \rightarrow 6\ 5\ 3\ 1 \rightarrow 5\ 4\ 4\ 2 \rightarrow 4\ 4\ 3\ 3\ 1 \rightarrow 5\ 3\ 3\ 2\ 2 \rightarrow 5\ 4\ 2\ 2\ 1\ 1 \rightarrow 6\ 4\ 3\ 1\ 1 \rightarrow 5\ 5\ 3\ 2 \rightarrow 4\ 4\ 4\ 2\ 1 \rightarrow 5\ 3\ 3\ 3\ 1 \rightarrow 5\ 4\ 2\ 2\ 2 \rightarrow 5\ 4\ 3\ 1\ 1\ 1 \rightarrow 6\ 4\ 3\ 2 \rightarrow 5\ 4\ 3\ 2\ 1$. From 4 4 3 2 1 1 it is 20 moves, the most possible with 15 counters: $4\ 4\ 3\ 2\ 1\ 1 \rightarrow 6\ 3\ 3\ 2\ 1 \rightarrow 5\ 5\ 2\ 2\ 1 \rightarrow 5\ 4\ 4\ 1\ 1 \rightarrow 5\ 4\ 3\ 3 \rightarrow 4\ 4\ 3\ 2\ 2 \rightarrow 5\ 3\ 3\ 2\ 1\ 1 \rightarrow 6\ 4\ 2\ 2\ 1 \rightarrow 5\ 5\ 3\ 1\ 1 \rightarrow 5\ 4\ 4\ 2 \rightarrow 4\ 4\ 3\ 3\ 1 \rightarrow 5\ 3\ 3\ 2\ 2 \rightarrow 5\ 4\ 2\ 2\ 1\ 1 \rightarrow 6\ 4\ 3\ 1\ 1 \rightarrow 5\ 5\ 3\ 2 \rightarrow 4\ 4\ 4\ 2\ 1 \rightarrow 5\ 3\ 3\ 3\ 1 \rightarrow 5\ 4\ 2\ 2\ 2 \rightarrow 5\ 4\ 3\ 1\ 1\ 1 \rightarrow 6\ 4\ 3\ 2 \rightarrow 5\ 4\ 3\ 2\ 1$. The two runs merge at 5 4 4 2.

Hints, grades 4–5, in the order to give them

1. Wrong rows (usually a forgotten pile): “The new pile equals the number of piles before the move. Check that first.”
2. Problem 6, to speed up: “Which six arrangements are in the two loops you found? Look for them in your rows, and stop as soon as one appears.” Then “If your run reaches an arrangement someone else has already finished, stop; you know where it ends.”
3. Problem 7: “How many rows have you written by move 7? How many different arrangements exist?”
4. Problem 10: “After the move, which pile is the new one? Could it be smaller than some other pile?” then “Cover the biggest pile with your hand. What is left, before and after?”
5. Problem 12: “The new pile tells you how many piles there were. Choose which pile is new.”
6. Problem 13: “Compare the biggest pile with the number of piles in each unreachable one.”
7. Problem 14: “The slowest start for 6 counters was 2 2 1 1: small piles, but not all ones. Try something like that with 10.”
8. Problem 15: “Write the loop for 7 counters as stairs 3 2 1 0 with one extra counter each, and watch where the extra counter goes at each move.”

If the grades 4–5 table stalls or finishes early

Stalls: Problem 6 can be cut to the seven S starts (give them the list and ask them to check) and Problem 9 to the table only. If the arguments in Problems 10 and 13 do not come, do them as a conversation with counters on the table: make a move, cover the new pile, ask what is left.

Finishes early: (1) 21 counters in one pile takes 15 moves ($21 \rightarrow 20\ 1 \rightarrow 19\ 2 \rightarrow 18\ 2\ 1 \rightarrow 17\ 3\ 1 \rightarrow 16\ 3\ 2 \rightarrow 15\ 3\ 2\ 1 \rightarrow 14\ 4\ 2\ 1 \rightarrow 13\ 4\ 3\ 1 \rightarrow 12\ 4\ 3\ 2 \rightarrow 11\ 4\ 3\ 2\ 1 \rightarrow 10\ 5\ 3\ 2\ 1 \rightarrow 9\ 5\ 4\ 2\ 1 \rightarrow 8\ 5\ 4\ 3\ 1 \rightarrow 7\ 5\ 4\ 3\ 2 \rightarrow 6\ 5\ 4\ 3\ 2\ 1$); from 5 5 4 3 2 1 1 it takes 30, the most possible. (2) Predict the loops for 9 counters (the 3–2–1 stairs with extras on three of the four steps: 4 3 2, 4 3 1 1, 4 2 2 1, 3 3 2 1, one loop of four) and for 13 (stairs 4–3–2–1 with extras on three of five steps: ten arrangements, two loops of five). (3) Count the Garden of Eden arrangements for 6, 7, 8 counters using the rule from Problem 13 (3, 5, 7) and check against the lists.

Share (5 minutes)

Each table shows one thing, with counters, in this order. K–1: the three stairs they built (3, 6, 10) and the claim “these numbers stop; 4, 5, 7 go round and round.” Grades 2–3: the map of 6 counters, held up,

with the sentence “every arrow chain ends at the stairs,” and the 8-counter surprise (two different loops). Grades 4–5: why no arrangement of 8 counters can stand still, in the three steps of Problem 10, and the prediction for 11 counters. Close with the question nobody has answered yet: “we checked every arrangement of 6 counters by hand, and a computer checked all 42 arrangements of 10. How would you know it for 15, 21, 28, without checking?” That is the theorem.

The mathematics behind the session (for the two mathematicians)

Setting

A state is a partition of n ; write it as towers of dots, columns of heights $\lambda_1 \geq \lambda_2 \geq \dots \geq \lambda_m$ standing on a floor. The move B removes the bottom row (one dot from each of the m columns) and stands it up as a new column of height m ; then the columns are re-sorted by height. The state space is finite and B is a function, so every orbit is eventually periodic: this is the pigeonhole argument of grades 4–5 Problem 7, and the functional graph (the “map” of grades 2–3 Problem 13) consists of cycles with trees hanging off them.

Fixed points. If $B(\lambda) = \lambda$ then the new column, of height m , must be the tallest column afterwards (every other column shrank), so $\lambda_1 = m$. Removing that column from both sides, the multiset $\{\lambda_1 - 1, \lambda_2 - 1, \dots, \lambda_m - 1\}$ with zeros dropped must equal $\{\lambda_2, \dots, \lambda_m\}$; comparing largest elements repeatedly gives $\lambda_2 = \lambda_1 - 1$, $\lambda_3 = \lambda_2 - 1$, and so on down to $\lambda_m = 1$. So the fixed points are exactly the staircases $(k, k - 1, \dots, 1)$, and only the triangular numbers $T_k = k(k + 1)/2$ admit one (grades 4–5 Problem 10).

Brandt’s theorem and the structure of the cycles

Theorem (Brandt 1982). If $n = T_k$, every state reaches the staircase. More precisely, for every n with $T_k \leq n < T_{k+1}$, write $n = T_k + r$ with $0 \leq r \leq k$; the states that lie on cycles are exactly the staircase $(k, \dots, 1)$ with one extra dot added to each of r distinct “steps” (a step being one of the k columns or a new column of height 1), and B moves the set of occupied steps cyclically: if the extras occupy steps $S \subseteq \{0, 1, \dots, k\}$ (step j meaning column of height j gets one more; step 0 meaning a new column of height 1), then after a move they occupy $S - 1$ modulo $k + 1$. Hence the cycle lengths divide $k + 1$, the number of cycle states is $\binom{k+1}{r}$, and the number of cycles is the number of binary necklaces of length $k + 1$ with r ones. For $n = T_k$ ($r = 0$) the only cycle is the fixed point.

Checks against the session: $n = 8 = T_3 + 2$, necklaces of length 4 with two ones: 1100 (period 4) and 1010 (period 2), the loops L and S; $n = 12 = T_4 + 2$, necklaces of length 5 with two ones: two of them, both of period 5; $n = 7, 9, 11, 14$: one cycle each; $n = 17 = T_5 + 2$: three cycles, of lengths 6, 6 and 3.

Proof sketch (a potential function plus a rotation argument). Number the cells of a state by row r (from the floor, $r = 1$ at the bottom) and column c (from the left). Perform the move in two steps. Step 1: remove the bottom row and put it as a new column at the far left, shifting the old columns one place right; a surviving cell goes from (r, c) to $(r - 1, c + 1)$ and a bottom cell from $(1, c)$ to $(c, 1)$. Every cell keeps its diagonal number $d = r + c$. Step 2: if the new column list $(m, \lambda_1 - 1, \dots, \lambda_m - 1)$ is not in descending order, sort it. Let $W(\lambda) = \sum_{\text{cells}} (r + c)$. Step 1 keeps W ; step 2 permutes column heights, and $\sum_c c \cdot h_c$ is minimised, strictly, by sorting the heights in descending order (rearrangement inequality), so W strictly decreases whenever sorting is needed. Since W is a positive integer, sorting happens only finitely often along any orbit; in particular, on a cycle, no move ever needs sorting, and every cell simply travels along its diagonal: on diagonal d the $d - 1$ positions $(1, d - 1), (2, d - 2), \dots, (d - 1, 1)$ are visited cyclically, with period $d - 1$.

Now suppose a cycle state has a hole (an empty position) on diagonal d while some cell sits on diagonal

$d + 1$. The hole rotates with period $d - 1$ and the cell with period d ; these are coprime, so at some time in the cycle the hole and the cell are in the same row, the cell at $(r, d + 1 - r)$ immediately to the right of the empty $(r, d - r)$. That contradicts descending column heights. Hence on a cycle the diagonals $2, 3, \dots$ are full, full, $\dots$, full, then possibly one partially filled diagonal, then empty. The full diagonals form a staircase T_k , the partial diagonal $k + 2$ carries the r extras on distinct steps, and the rotation is the cyclic shift described above. If $n = T_k$ the partial diagonal is empty and the cycle is the fixed point. (Compare Akin and Davis, 1985.)

How many moves

Igusa (1985) and Etienne (1991) proved that for $n = T_k$ every start reaches the staircase within $k^2 - k$ moves. The bound is attained by $(k - 1, k - 1, k - 2, \dots, 2, 1, 1)$ for $k = 3, 4, 5, 6$: 6 moves from 2 2 1 1, 12 from 3 3 2 1 1, 20 from 4 4 3 2 1 1, 30 from 5 5 4 3 2 1 1 (every fact in this paragraph was checked by exhaustive computation while preparing this guide). For 10 counters the distribution of the number of moves over the 42 arrangements is: 0 moves, 1 arrangement; 1, 1; 2, 3; 3, 5; 4, 5; 5, 3; 6, 4; 7, 4; 8, 4; 9, 3; 10, 3; 11, 3; 12, 3.

Garden of Eden states

A state with m piles has a predecessor if and only if its largest pile is at least $m - 1$: choosing a pile of size $p \geq m - 1$ as the new pile, the other $m - 1$ piles came from piles one larger, and $p - (m - 1) \geq 0$ piles of size 1 complete the predecessor. So the Garden of Eden states are the partitions with largest part at most (number of parts) $- 2$; by conjugation these are counted by partitions of n with largest part minus number of parts at least 2, that is, Dyson rank at least 2, which is the route by which Hopkins and Sellers (2007) enumerated them. Counts for $n = 3, \dots, 15$: 1, 1, 2, 3, 5, 7, 10, 14, 20, 27, 37, 49, 66. Undoing a move (grades 4–5 Problem 12) is the same computation read backwards; a state can have several predecessors (3 2 1 has two, 5 3 2 has three).

Where it leads

Undergraduate: partitions and Young diagrams; functional graphs and eventually periodic dynamics on finite sets; the rearrangement inequality as a potential-function tool; necklaces and Burnside counting for the number of cycles; Dyson's rank. Research: the transient length and the shape of the trees (Griggs and Ho 1998 study the analogous shift on compositions and other "shift" dynamics); generalisations such as Austrian, Carolina and Montreal solitaire and σ -shifts on partitions (surveyed by Hopkins 2012); random Bulgarian solitaire, where each pile contributes a counter with probability p , studied as a Markov chain; the exact enumeration of Garden of Eden partitions and the distribution of the number of moves.

References. J. Brandt, Cycles of partitions, Proc. Amer. Math. Soc. 85 (1982). M. Gardner, Mathematical Games, Scientific American, August 1983 (the column that made the game known). E. Akin and M. Davis, Bulgarian solitaire, Amer. Math. Monthly 92 (1985). K. Igusa, Solution of the Bulgarian solitaire conjecture, Math. Magazine 58 (1985). G. Etienne, Tableaux de Young et solitaire bulgare, J. Combin. Theory Ser. A 58 (1991). J. R. Griggs and C.-C. Ho, The cycling of partitions and compositions under repeated shifts, Adv. in Appl. Math. 21 (1998). B. Hopkins and J. A. Sellers, Exact enumeration of Garden of Eden partitions, Integers 7(2) (2007). B. Hopkins, 30 years of Bulgarian solitaire, College Math. J. 43 (2012).

Tables: every arrangement, its successor, and its distance

Each table lists every arrangement (Start), the arrangement after one move (Next), the number of moves until the game stops or first enters its loop (Moves), and, for 8 counters, which loop it enters (L: the loop of four, S: the loop of two). A star in the GoE column marks a Garden of Eden arrangement, one that no move produces.

6 counters, 11 arrangements			
Start	Next	Moves	GoE
6	5 1	3	
5 1	4 2	2	
4 2	3 2 1	1	
4 1 1	3 3	5	
3 3	2 2 2	4	
3 2 1	3 2 1	0	
3 1 1 1	4 2	2	
2 2 2	3 1 1 1	3	
2 2 1 1	4 1 1	6	*
2 1 1 1 1	5 1	3	*
1 1 1 1 1 1	6	4	*

7 counters, 15 arrangements			
Start	Next	Moves	GoE
7	6 1	3	
6 1	5 2	2	
5 2	4 2 1	1	
5 1 1	4 3	2	
4 3	3 2 2	1	
4 2 1	3 3 1	0	
4 1 1 1	4 3	2	
3 3 1	3 2 2	0	
3 2 2	3 2 1 1	0	
3 2 1 1	4 2 1	0	
3 1 1 1 1	5 2	2	*
2 2 2 1	4 1 1 1	3	*
2 2 1 1 1	5 1 1	3	*
2 1 1 1 1 1	6 1	3	*
1 1 1 1 1 1 1	7	4	*

8 counters, 22 arrangements				
Start	Next	Moves	Loop	GoE
8	7 1	4	L	
7 1	6 2	3	L	
6 2	5 2 1	2	L	
6 1 1	5 3	2	S	
5 3	4 2 2	1	S	
5 2 1	4 3 1	1	L	
5 1 1 1	4 4	2	L	
4 4	3 3 2	1	L	
4 3 1	3 3 2	0	L	
4 2 2	3 3 1 1	0	S	
4 2 1 1	4 3 1	0	L	
4 1 1 1 1	5 3	2	S	
3 3 2	3 2 2 1	0	L	
3 3 1 1	4 2 2	0	S	
3 2 2 1	4 2 1 1	0	L	
3 2 1 1 1	5 2 1	2	L	*
3 1 1 1 1 1	6 2	3	L	*
2 2 2 2	4 1 1 1 1	3	S	*
2 2 2 1 1	5 1 1 1	3	L	*
2 2 1 1 1 1	6 1 1	3	S	*
2 1 1 1 1 1 1	7 1	4	L	*
1 1 1 1 1 1 1 1	8	5	L	*

9 counters, 30 arrangements (1 of 2)			
Start	Next	Moves	GoE
9	8 1	5	
8 1	7 2	4	
7 2	6 2 1	3	
7 1 1	6 3	3	
6 3	5 2 2	2	
6 2 1	5 3 1	2	
6 1 1 1	5 4	2	
5 4	4 3 2	1	
5 3 1	4 3 2	1	
5 2 2	4 3 1 1	1	
5 2 1 1	4 4 1	6	
5 1 1 1 1	5 4	2	
4 4 1	3 3 3	5	
4 3 2	3 3 2 1	0	
4 3 1 1	4 3 2	0	

9 counters, 30 arrangements (2 of 2)

Start	Next	Moves	GoE
4 2 2 1	4 3 1 1	0	
4 2 1 1 1	5 3 1	2	
4 1 1 1 1 1	6 3	3	*
3 3 3	3 2 2 2	4	
3 3 2 1	4 2 2 1	0	
3 3 1 1 1	5 2 2	2	*
3 2 2 2	4 2 1 1 1	3	
3 2 2 1 1	5 2 1 1	7	*
3 2 1 1 1 1	6 2 1	3	*
3 1 1 1 1 1 1	7 2	4	*
2 2 2 2 1	5 1 1 1 1	3	*
2 2 2 1 1 1	6 1 1 1	3	*
2 2 1 1 1 1 1	7 1 1	4	*
2 1 1 1 1 1 1 1	8 1	5	*
1 1 1 1 1 1 1 1 1	9	6	*

10 counters, 42 arrangements (1 of 2)

Start	Next	Moves	GoE
10	9 1	6	
9 1	8 2	5	
8 2	7 2 1	4	
8 1 1	7 3	9	
7 3	6 2 2	8	
7 2 1	6 3 1	3	
7 1 1 1	6 4	3	
6 4	5 3 2	2	
6 3 1	5 3 2	2	
6 2 2	5 3 1 1	7	
6 2 1 1	5 4 1	11	
6 1 1 1 1	5 5	7	
5 5	4 4 2	6	
5 4 1	4 3 3	10	
5 3 2	4 3 2 1	1	
5 3 1 1	4 4 2	6	
5 2 2 1	4 4 1 1	11	
5 2 1 1 1	5 4 1	11	
5 1 1 1 1 1	6 4	3	
4 4 2	3 3 3 1	5	
4 4 1 1	4 3 3	10	

10 counters, 42 arrangements (2 of 2)

Start	Next	Moves	GoE
4 3 3	3 3 2 2	9	
4 3 2 1	4 3 2 1	0	
4 3 1 1 1	5 3 2	2	
4 2 2 2	4 3 1 1 1	3	
4 2 2 1 1	5 3 1 1	7	
4 2 1 1 1 1	6 3 1	3	*
4 1 1 1 1 1 1	7 3	9	*
3 3 3 1	4 2 2 2	4	
3 3 2 2	4 2 2 1 1	8	
3 3 2 1 1	5 2 2 1	12	*
3 3 1 1 1 1	6 2 2	8	*
3 2 2 2 1	5 2 1 1 1	12	*
3 2 2 1 1 1	6 2 1 1	12	*
3 2 1 1 1 1 1	7 2 1	4	*
3 1 1 1 1 1 1 1	8 2	5	*
2 2 2 2 2	5 1 1 1 1 1	4	*
2 2 2 2 1 1	6 1 1 1 1	8	*
2 2 2 1 1 1 1	7 1 1 1	4	*
2 2 1 1 1 1 1 1	8 1 1	10	*
2 1 1 1 1 1 1 1 1	9 1	6	*
1 1 1 1 1 1 1 1 1 1	10	7	*